

AUTOMATED PODCAST TRANSCRIPTION AND TOPIC SEGMENTATION



presented by: Dharani

Problem Statement & Solution

- ▶ Problem:
- ▶ Manual transcription and analysis of podcast audio is time-consuming and error-prone.
- ▶ Solution:
- ▶ Automated system to clean audio, generate transcripts, segment topics, extract keywords, and visualize insights using a dashboard.

Project Objectives

- ▶ - Automate podcast transcription
- ▶ - Clean raw audio files
- ▶ - Segment text into meaningful topics
- ▶ - Extract keywords
- ▶ - Visualize data using charts and word clouds
- ▶ - Provide user-friendly dashboard

System Architecture

- ▶ Audio Input → Audio Cleaning → Transcription → Preprocessing → Segmentation → Keyword Extraction → Visualization → UI Dashboard

Technologies Used

- ▶ Python
- ▶ Streamlit
- ▶ Matplotlib
- ▶ WordCloud
- ▶ OS & File Handling
- ▶ NLP Libraries

Usage of UI

- ▶ Upload transcript file
- ▶ View transcription
- ▶ View segmented text
- ▶ View extracted keywords
- ▶ View visualizations
- ▶ Play cleaned audio

Limitations & Challenges

- ▶ - Large audio files take more processing time
- ▶ - Keyword extraction is basic
- ▶ - Requires manual folder setup
- ▶ - Limited scalability

Future Scope

- ▶ - Real-time transcription
- ▶ - Sentiment analysis
- ▶ - Speaker identification
- ▶ - Cloud deployment
- ▶ - Multi-language support

Thank You

- ▶ Thank you for your attention!